

## TREMORTRACE · DESIGN PRECIS

# TremorTrace design precis

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## DOCUMENT CONTROL

|             |                                          |         |                |
|-------------|------------------------------------------|---------|----------------|
| Document ID | TRT-PRC-001                              | Version | 0.5            |
| Status      | <b>Draft</b>                             | Date    | 2026-09-25     |
| Author      | Amish Chadha                             | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept            |         |                |
| Repository  | github.com/BoujeeEnjinia1701/tremortrace |         |                |
| Source      | docs/02-concept.md @ 5a589c0             |         |                |

## REVISION HISTORY

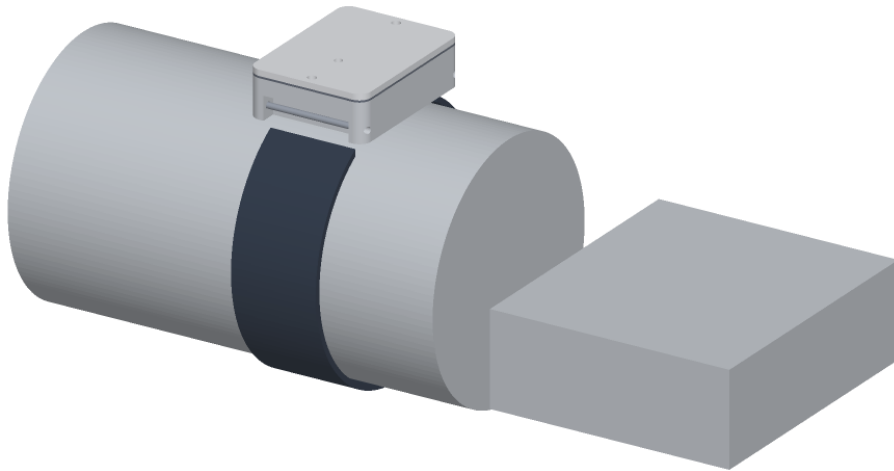
| Version | Date       | Author       | Change                                                                                                                                                                         |
|---------|------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.5     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002)                                                                                                                                    |
| 0.4     | 2026-09-25 | Amish Chadha | TRL 3: decisions from TRT-DDR-001 recorded; numbers from TRT-CAL-001 (storage ratio 390, not 1,000; firmware in internal flash; mass 29.4 g); parametric model and TRT-DWG-002 |
| 0.3     | 2026-09-24 | Amish Chadha | Review fixes: one-module and local-first choices marked proposed; massing pod closed at 12 mm; concept sheet renumbered TRT-DWG-001                                            |
| 0.2     | 2026-09-24 | Amish Chadha | Populate to TRL 2 (architecture, first-order numbers, safety, media)                                                                                                           |
| 0.1     | 2026-09-24 | Amish Chadha | Initial scaffold                                                                                                                                                               |

## TremorTrace design precis

TremorTrace is a small wrist pod on a watch strap that records wrist motion all day, turns it into 10-second tremor summaries on the band, and hands them to an open notebook that draws a daily tremor profile. The TRL 3 calculations (TRT-CAL-001) show that one off-the-shelf module and a 150 mAh cell meet nine of eleven requirements for \$43.19 in parts, including mass at 25.4 g against a 30 g limit with a textile strap; splash resistance and the notebook can only be verified with hardware and software work.

### TremorTrace

CONCEPT, NOT FOR FABRICATION



Grey: Forearm and hand for scale

### ■ How it works

1. **Sense.** A 6-axis IMU (accelerometer and gyroscope) samples wrist motion at 104 Hz, with ranges of  $\pm 8$  g and  $\pm 2000$  dps.
2. **Summarize on the band.** Every 10 seconds the firmware band-pass filters the signal to 3 to 15 Hz, runs a spectrum, and stores a 32-byte summary: dominant frequency, band RMS of acceleration and angular velocity, total RMS, a band-to-low-frequency power ratio and an activity flag. Tremor values are kept even when the flag says the wearer was moving, so tremor during activity is not lost.
3. **Store.** Summaries go to the module's 2 MB flash, enough for 9.7 days at 16 h of wear per day without a phone. A 256 kB area holds up to 21 raw 10 s snippets captured on demand for research.

4. **Sync.** When the owner opens the web Bluetooth page or a laptop script, summaries transfer over Bluetooth Low Energy. Nothing goes to the cloud.
5. **Analyze.** An open Python notebook turns summaries into a daily profile: minutes of tremor per hour, dominant frequency and amplitude over the day, and optional medication-time markers entered by the owner.

**TremorTrace: data flow (local-first, no cloud)**

CONCEPT, NOT FOR FABRICATION



## ■ Main components

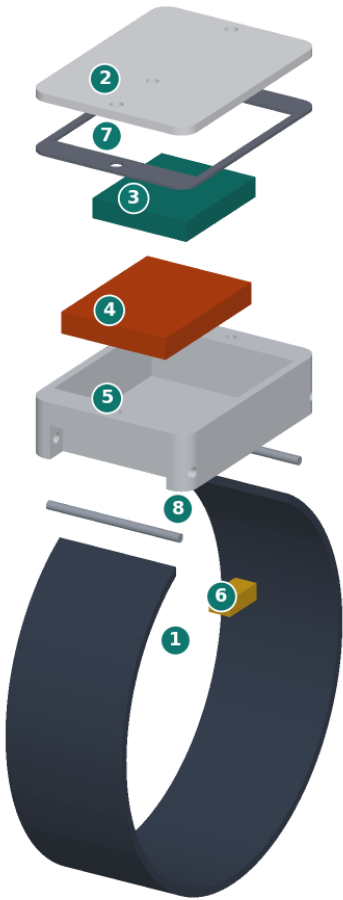
Table 1. Main components; item numbers match `bom/bom.csv` and the exploded view.

| # | Component          | Choice                                                                                  | Notes                                                                                                                                        |
|---|--------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Strap              | 22 mm woven textile two-piece quick-release strap                                       | Skin-safe and washable; the heaviest part (about 8 g, assumed). Replaced the 12 g silicone strap; decided by Amish, 2026-09-25 (TRT-DDR-002) |
| 2 | Enclosure lid      | 3D-printed PETG, 1.5 mm, two M2 screws                                                  | 2 mm light pipe over the module's status LED                                                                                                 |
| 3 | Controller and IMU | Seeed XIAO nRF52840 Sense class: nRF52840, LSM6DS3TR-C IMU, 2 MB flash, BQ25101 charger | One module, no custom PCB. Decided by Amish, 2026-09-25 (TRT-DDR-001)                                                                        |
| 4 | Battery            | 150 mAh protected LiPo, 19.75 x 26.02 x 3.8 mm (Adafruit 1317 class)                    | See power budget                                                                                                                             |
| 5 | Enclosure base     | 3D-printed PETG with integrated 22 mm lugs                                              | Charging connector through the floor                                                                                                         |
| 6 | Charging contacts  | Magnetic 2-pin pogo connector                                                           | Keeps the enclosure sealed                                                                                                                   |
| 7 | Gasket             | Printed TPU ring, 0.5 mm compressed                                                     | Toward the IP54 target (R7)                                                                                                                  |
| 8 | Spring bars        | 22 mm                                                                                   | Hold the strap in the lugs                                                                                                                   |

**TremorTrace: exploded view**

CONCEPT, NOT FOR FABRICATION

- 1 Textile strap
- 2 Enclosure lid
- 3 Controller and IMU module
- 4 LiPo cell, 150 mAh
- 5 Enclosure base
- 6 Charging connector
- 7 TPU gasket
- 8 Spring bars



■ **Key numbers**

All values come from TRT-CAL-001 and the parametric model in `cad/src/model.py`.

Table 2. Key numbers at TRL 3.

| Quantity             | Value                                      | Requirement                                  |
|----------------------|--------------------------------------------|----------------------------------------------|
| Frequency resolution | 0.100 Hz bins (10 s at 104 Hz)             | R1 (0.2 Hz) met, with sample-rate correction |
| Nyquist limit        | 52 Hz                                      | R1, R3 met                                   |
| In-band noise        | 0.31 mg and 0.017 dps RMS                  | R2 met                                       |
| Recording current    | 1.12 mA nominal, 1.54 mA conservative      |                                              |
| Battery life         | 6.3 days nominal, 4.5 days conservative    | R4 (3 days) met, if the firmware sleeps      |
| Summary storage      | 184.3 kB per day (32 bytes per 10 s, 16 h) |                                              |
| On-band history      | 9.7 days                                   | R5 (7 days) met                              |
| Pod size             | 40 x 30 x 12 mm                            | R6 size met                                  |
| Mass                 | 25.4 g with textile strap                  | R6 (30 g) met, 4.6 g margin                  |
| Parts cost           | \$43.19                                    | R10 met                                      |

The general arrangement is drawing TRT-DWG-002 ( [cad/drawings/TRT-DWG-002.pdf](#) ).

## ■ Key design choices

- **Summaries, with optional raw snippets.** A 32-byte summary is 390 times smaller than the 12,480 bytes of raw data in a 10 s window, and it keeps personal data minimal. Decided by Amish, 2026-09-25 (TRT-DDR-001).
- **One module, no custom PCB.** Keeps the first build within reach of anyone with a soldering iron and a 3D printer. Decided by Amish, 2026-09-25 (TRT-DDR-001).
- **Web Bluetooth page first.** No app store and no account; a native app may follow. Decided by Amish, 2026-09-25 (TRT-DDR-001).
- **Local-first data.** No cloud account is needed to use or analyze the data (R9). Decided by Amish, 2026-09-25 (TRT-DDR-001).
- **Most affected wrist first.** Decided by Amish, 2026-09-25 (TRT-DDR-001).
- **Charge current 50 mA.** Charges in about 3.6 h with about 65 mW of charger heat, against 1.8 h and 130 mW at 100 mA. The firmware selects the 50 mA setting. Decided by Amish, 2026-09-25 (TRT-DDR-002).
- **Textile strap.** A woven textile strap of about 8 g replaces the 12 g silicone strap and raises the R6 mass margin from 0.6 g to 4.6 g. Decided by Amish, 2026-09-25 (TRT-DDR-002).

## ■ Safety

**Safety:** Research and educational prototype only. It is not a medical device, has not been cleared or approved by any regulator, and must not be used to diagnose, treat or adjust medication.

- The LiPo cell sits against the skin: use a protected cell, never charge while worn, charge at no more than the cell's 150 mA limit, and stop use if the pod becomes warm or swollen.
- Use skin-safe strap materials and check for irritation during long wear.
- Tremor data is health data. Keep it on the owner's devices and share only with consent.

## ■ Open questions

- The activity flag marks tremor during a reach as activity in synthetic data (TRT-CAL-001, section 4). How well can action tremor be separated from voluntary movement with 10 s summaries?
- Confirm the IMU sample-rate tolerance, the module's run current and the textile strap mass.
- First clinical or patient partner for co-design and eventual validation. Proposed, awaiting Amish.

Concept media: [blueprint sheet](#), [interactive 3D model](#).